$\langle\psi| B A |\psi\rangle$ $\qquad$ $(A|\psi\rangle)^\dagger = \langle\psi| A^\dagger$

$$T(\hat{\phi}(x)\,\hat{\phi}(y)) = \underbrace{\theta(x^0-y^0)}\,\hat{\phi}(y)\,\hat{\phi}(x) + \theta(y^0-x^0)\,\hat{\phi}(x)\,\phi(x)$$

↳ Heviside step fn $\qquad \theta(s) = \begin{cases} 1 & s \geq 0 \\ 0 & s < 0 \end{cases}$

$$\langle 0| A \left(\bar{\psi}(x)\,\psi(y)\,|0\rangle\right)$$

$$\langle 0|(A\bar{\psi})\,\psi\,|0\rangle$$

Green's function $\quad D$ linear differential operator $\quad$ diff. eq. $\quad D u(x) = 0$

ex: $\quad (\partial_t^2 + \partial_x^2)\, u(x,y) = 0$

Green funct. associated to D is the solution to

$D_x G(x, x') = \delta(x - x')$ $\quad \leftarrow$ in the sense of distributions

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to solve $\quad D u(x) = f(x) \quad$ we use G:

$$f(x) = \int ds\ \delta(x-s)\, f(s) = \int ds\ D_x G(x,s)\, f(s) = D_x \int ds\ G(x,s)\, f(s)$$

$u(x) = \int ds\ G(x,s)\, f(s) \quad$ is a solution

If D is translation invariant $\left(D_x u(x+s) = D_{x+s}\, u(x+s) \right)$

then $\quad G(x,x') = G(x - x') \quad$ and $\quad \boxed{D\, G(x) = \delta(x)}$

Fourier transform: $\quad \hat{D}\hat{G}(p) = 1 \quad \rightarrow \quad \hat{G}(p)$ inverse of $\hat{D}$

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ex: $\quad D = \frac{d}{dx} \quad \hat{D} = ip \quad \rightarrow \quad ip\, \hat{G}(p) = 1 \Rightarrow \hat{G}(p) = \frac{-i}{p}$

$$\lim_{\varepsilon \to 0^+} \int dx\ \frac{-i}{p + i\varepsilon}\, e^{-ipx}$$

$$\int_{-\infty}^{\infty} \frac{1}{x}\, dx = \lim_{\varepsilon \to 0} \left(\int_{-\infty}^{\varepsilon} \frac{1}{x}\, dx + \int_{\varepsilon}^{\infty} \frac{1}{x}\, dx \right)$$

$$= \lim_{\varepsilon \to 0} \left(-\int_{\varepsilon}^{-\infty} \frac{1}{x}\, dx + \int_{\varepsilon}^{\infty} \frac{1}{x}\, dx \right)$$

$$= \lim_{\varepsilon \to 0} \left(-\int_{-\varepsilon}^{\infty} \frac{1}{x}\, dx + \int_{\varepsilon}^{\infty} \frac{1}{x}\, dx \right) = ?$$

principal value

$\langle \text{P.V.} \frac{1}{x}, f \rangle \quad$ we want something like $\int_{-\infty}^{\infty} \frac{1}{x} f(x)\, dx$

$$\langle \text{P.v.}\tfrac{1}{x}, f\rangle = \lim_{\varepsilon\to 0^+}\left[\int_{-\infty}^{-\varepsilon}\frac{1}{x}f(x)\,dx - \int_{\varepsilon}^{\infty}\frac{1}{x}f(x)\,dx\right]$$

$$\langle \text{P.v.}\tfrac{1}{x}, 1\rangle = \lim_{\varepsilon\to 0^+}\left[\int_{-\infty}^{-\varepsilon}\frac{1}{x}\,dx - \int_{\varepsilon}^{\infty}\frac{1}{x}\,dx\right]$$

$$= \lim_{\varepsilon\to 0^+}\left[-\int_{\infty}^{\varepsilon}\frac{1}{x}\,dx - \int_{\varepsilon}^{\infty}\frac{1}{x}\,dx\right] = 0$$

Fourier transform of $\frac{1}{x} = -i\pi\,\mathrm{sgn}(p)$

Fourier transform of $\frac{1}{p} = \frac{i}{2}\,\mathrm{sgn}(x)$

$$\int \frac{-i}{p} e^{ipx}\,dp$$

$$\hat{G}(p) = \frac{-i}{p} \qquad G(x) = \frac{1}{2}\,\mathrm{sgn}(x) = \begin{cases} 1 & \text{if } x>0 \\ 0 & \text{if } x=0 \\ -1 & \text{if } x<0 \end{cases}$$

$$\frac{d}{dx}\frac{1}{2}\,\mathrm{sgn}(x) = \qquad \mathrm{sgn}(x) = \theta(x) - \theta(-x)$$

$$D = \frac{d}{dx} \qquad G(x) = \theta(x)$$

<u>Distributions</u> are linear functional that act on set of test functions

Physicist delta: $\int_{\mathbb{R}} \delta(x-x_0)\, f(x) = f(x_0)$ no such thing

δ_{x_0} is a map $\delta_{x_0} : S \to \mathbb{R}$

$f \mapsto f(x_0)$

$\delta_{x_0}[f]$

$\equiv$

$\langle \delta_{x_0}, f\rangle$

in general distribution is a map $T : S \to \mathbb{R}$

<u>Regular distributions</u> if f is integrable $(f \in L^1(\mathbb{R}))$

associate to it T_f defined by $\langle T_f, \phi\rangle = \int_{\mathbb{R}} f(x)\,\phi(x)\,dx$

distributions can be used to generalize diff. equation to things that are not continuous.

Distributional derivatives

if T_f is a regular distribution with $f \in S$ then

$$\int_{\mathbb{R}} f'(x)\,\phi(x)\,dx = -\int_{\mathbb{R}} f(x)\,\phi'(x)\,dx$$ integration by parts

$$\Rightarrow \langle T_{f'}, \phi \rangle = -\langle T_f, \phi' \rangle$$

in general if $T : S \to \mathbb{R}$ define T' by $\langle T', \phi \rangle = -\langle T, \phi' \rangle$

if $T = T_f$ with $f \in S$ then $(T_f)' = T_{f'}$

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if f is discontinuous $T_{f'}$ does not make sense but $(T_f)'$ does

ex: $f(x) = \theta(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$

I want to do $\theta'(x)$ but this is not defined at $x=0$

what we can do is look at $(T_\theta)'$

$$\langle T_\theta', \phi \rangle = -\langle T_\theta, \phi' \rangle = -\int_{-\infty}^{\infty} \theta(x)\,\phi'(x)\,dx$$

$$= -\int_0^{\infty} \phi'(x)\,dx$$

$$= -\left[\phi(x)\right]_0^{\infty} = -\underbrace{\lim_{x\to\infty} \phi(x)}_{=0} + \phi(0)$$

$$= \phi(0)$$

$$= \langle \delta_0, \phi \rangle$$

$$T_\theta' = \delta_0$$

$$\frac{d}{dx}\theta(x) = \delta(x)$$

$$G(x) = \frac{1}{2}\,\mathrm{sgn}(x) = \begin{cases} \frac{1}{2} & x > 0 \\ 0 & x = 0 \\ -\frac{1}{2} & x < 0 \end{cases}$$

$$\frac{d}{dx} G(x) = \delta(x)$$

$$\langle T_G', \phi \rangle = -\langle T_G, \phi' \rangle = -\int_{-\infty}^{\infty} \frac{1}{2}\,\mathrm{sgn}(x)\,\phi'(x)\,dx$$

$$= \underbrace{-\int_{0}^{\infty} \frac{1}{2}\phi'(x)\,dx}_{\frac{1}{2}\phi(0)} + \underbrace{\int_{-\infty}^{0} \frac{1}{2}\phi'(x)\,dx}_{\frac{1}{2}\phi(0)}$$

$$= \phi(0) = \langle \delta_0, \phi \rangle$$

$f = \frac{1}{x}$ not a nice function $\rightarrow$ T_f not exist

associate to this the distribution $\mathrm{p.v.}\,\frac{1}{x}$

$$\langle \mathrm{p.v.}\,\tfrac{1}{x}, \phi \rangle = \lim_{\varepsilon \to 0^+} \int_{\mathbb{R} \setminus (-\varepsilon, \varepsilon)} \frac{1}{x}\phi(x)\,dx$$

You can tene Fourier transforms of distributions

$T : S \to \mathbb{R}$

$$\langle \hat{T}, \phi \rangle = \langle T, \hat{\phi} \rangle \quad \rightarrow \text{Fourier transform}$$

$$\langle \check{T}, \phi \rangle = \langle T, \check{\phi} \rangle \quad \rightarrow \text{inverse transform}$$

if $f \in S$

$$\langle T_{\hat{f}}, \phi \rangle = \int_{-\infty}^{\infty} \hat{f}(p)\,\phi(p)\,dp$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dp\, \phi(p) \int_{-\infty}^{\infty} f(x)\, e^{-ipx}\,dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx\, f(x) \underbrace{\int_{-\infty}^{\infty} dp\, \phi(p)\, e^{-ipx}}_{\hat{\phi}(x)}$$

$$= \int_{-\infty}^{\infty} f(x)\,\hat{\phi}(x)\,dx = \langle T_f, \hat{\phi} \rangle$$

Fourier transform of δ?

$$\langle \hat{\delta}_0, \phi \rangle = \langle \delta_0, \hat{\phi} \rangle = \hat{\phi}(0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(x)\, e^{0}\, dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(x)\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \langle T_{1}, \phi \rangle$$

$$\hat{\delta}(x) = \frac{1}{\sqrt{2\pi}}$$

$$\langle \check{\text{P.V.}}\tfrac{1}{p}, \phi \rangle = \langle \text{P.V.}\tfrac{1}{p}, \check{\phi} \rangle \overset{?}{=} \langle T_{\frac{1}{2}\operatorname{sgn}}, \phi \rangle$$

$ds = -du$

$u = -s$

$$\lim_{\varepsilon \to 0^+} \int_{\mathbb{R} \setminus (-\varepsilon, \varepsilon)} ds\, \frac{1}{s}\, \check{\phi}(s) = \lim_{\varepsilon \to 0^+} \left[\int_{\varepsilon}^{\infty} ds\, \frac{1}{s}\, \check{\phi}(s) + \int_{-\infty}^{-\varepsilon} ds\, \frac{1}{s}\, \check{\phi}(s) \right]$$

$$= \lim_{\varepsilon \to 0^+} \left[\int_{\varepsilon}^{\infty} ds\, \frac{1}{s}\, \check{\phi}(s) - \int_{\varepsilon}^{\infty} du\, \frac{1}{u}\, \check{\phi}(-u) \right]$$

$$= \lim_{\varepsilon \to 0^+} \lim_{\Lambda \to \infty} \int_{\varepsilon}^{\Lambda} ds\, \frac{\check{\phi}(s) - \check{\phi}(-s)}{s}$$

$$\check{\phi}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dp\, e^{ipx}\, \phi(p)$$

$$= \frac{1}{2\pi} \lim_{\varepsilon \to 0^+} \int_{\varepsilon}^{\infty} ds\, \frac{1}{s} \left[\int_{-\infty}^{\infty} dp\, e^{ips}\, \phi(p) - \int_{-\infty}^{\infty} dp\, e^{-ips}\, \phi(p) \right]$$

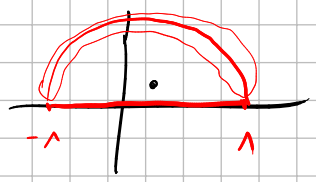
$$= \frac{1}{2\pi} \lim_{\varepsilon \to 0^+} \int_{\varepsilon}^{\infty} ds\, \frac{1}{s} \int_{-\infty}^{\infty} dp\, 2i \sin(ps)\, \phi(p)$$

need to use $\displaystyle\lim_{\Lambda \to \infty} \int_{0}^{\Lambda} \frac{\sin(\alpha x)}{x}\, dx = \frac{\pi}{2} \operatorname{sgn}(\alpha)$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} dp\, \phi(p) \lim_{\Lambda \to \infty} \int_{0}^{\Lambda} ds\, \frac{1}{s}\, 2i \sin(ps)$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} 2i\, \phi(p)\, \frac{\pi}{2} \operatorname{sgn}(p) = \frac{i}{2} \int_{-\infty}^{\infty} \phi(p) \operatorname{sgn}(p) = \langle T_{\frac{i}{2}\operatorname{sgn}}, \phi \rangle$$

$$G(x) = \frac{1}{2\pi} \lim_{\varepsilon \to 0^+} \lim_{\Lambda \to \infty} \int_{-\Lambda}^{\Lambda} dp \, e^{ipx} \frac{1}{p + i\varepsilon}$$



<u>Actual example</u>

$$D = \partial_t^2 - \nabla^2 \qquad \hat{D} = -p_0^2 + p_x^2 + p_y^2 + p_z^2$$

$$\hat{D}\,\hat{G}(p) = 1 \quad \Rightarrow \quad \hat{G}(p) = -\frac{1}{p^2}$$

$$p^2 \rightarrow \eta^{\mu\nu} p_\mu p_\nu$$

$$G(x) = \frac{-1}{(2\pi)^4} \int d^4p \, \frac{e^{ip\cdot x}}{p^2 + i\varepsilon}$$

$$p \cdot x \rightarrow \eta^{\mu\nu} x^\mu p_\nu$$

Feynman

$$-(p_0 + i\varepsilon)^2 + p_x^2 + p_y^2 + p_z^2$$